



Polynomial Long Division

If you know how to use long division with numbers, then you can use the same strategy to divide polynomials.

You can set up a division problem with polynomials the same way you do with numbers. Place the dividend inside the division symbol and the divisor to the left of it. The quotient goes on top of the division symbol.

	Numbers	Polynomials
Vertically	$\begin{array}{r} 5 \\ 15 \overline{)75} \\ -75 \\ \hline 0 \end{array}$	$\begin{array}{r} x^2 \\ 3 \overline{)3x^2} \\ -3x^2 \\ \hline 0 \end{array}$
Horizontally	$75 \div 15 = 5$	$3x^2 \div 3 = x^2$

Dividing a Second-Degree Polynomial by a First-Degree Binomial

As you do when you divide numbers, repeat the process of divide, multiply, and subtract until no more divisions can be made.

EXAMPLE 1

Divide.

A $(x^2 - 6x - 27) \div (x + 3)$

SOLUTION**Step 1**

$$\begin{array}{r}
 \overline{)x^2 - 6x - 27} \\
 \underline{-(x^2 + 3x)} \\
 -9x - 27
 \end{array}$$

Divide: $\frac{x^2}{x} = x$.

Multiply: $x \cdot (x + 3) = x^2 + 3x$.

Subtract: $(x^2 - 6x) - (x^2 + 3x) = -9x$.

Bring down -27 .

Step 2

$$\begin{array}{r}
 \overline{)x^2 - 6x - 27} \\
 \underline{-(x^2 + 3x)} \\
 -9x - 27 \\
 \underline{-(-9x - 27)} \\
 0
 \end{array}$$

Divide: $\frac{-9x}{x} = -9$.

Multiply: $-9 \cdot (x + 3) = -9x - 27$.

Subtract: $(-9x - 27) - (-9x - 27) = 0$.

So $x^2 - 6x - 27$ divided by $x + 3$ is $x - 9$.

B $2x^2 + 13x - 4$ by $2x - 1$

SOLUTION**Step 1**

$$\begin{array}{r}
 \overline{)2x^2 + 13x - 4} \\
 \underline{-(2x^2 - x)} \\
 14x - 4
 \end{array}$$

Divide: $\frac{2x^2}{2x} = x$.

Multiply: $x \cdot (2x - 1) = 2x^2 - x$.

Subtract: $(2x^2 + 13x) - (2x^2 - x) = 14x$.

Bring down -4 .

Step 2

$$\begin{array}{r}
 \overline{)2x^2 + 13x - 4} \\
 \underline{-(2x^2 - x)} \\
 14x - 4 \\
 \underline{-(14x - 7)} \\
 3
 \end{array}$$

Divide: $\frac{14x}{2x} = 7$.

Multiply: $7 \cdot (2x - 1) = 14x - 7$.

Subtract: $(14x - 4) - (14x - 7) = 3$.

So $2x^2 + 13x - 4$ divided by $2x - 1$ is $x + 7$ with remainder 3. ■

Representing and Checking a Quotient

Names of Expressions in a Polynomial Division

$$\begin{array}{r}
 \text{divisor} \longrightarrow 2x - 1 \overline{) 2x^2 + 13x - 4} \xleftarrow{\text{dividend}} 2x^2 + 13x - 4 \\
 \underline{-(2x^2 - x)} \xleftarrow{\text{quotient}} x + 1 \\
 14x - 4 \\
 \underline{-(14x - 7)} \\
 3 \xleftarrow{\text{remainder}} 3
 \end{array}$$

Different Ways to Express Related Multiplication and Division Facts

With numbers	With polynomials
$13 \div 5 = 2 \text{ R}3$	$(2x^2 + 13x - 4) \div (2x - 1) = x + 7 \text{ R}3$
$\frac{13}{5} = 2 + \frac{3}{5}$	$\frac{2x^2 + 13x - 4}{2x - 1} = x + 7 + \frac{3}{2x - 1}$

Because multiplication and division are inverse operations, you can check your answer to a division problem by multiplying.

Using Multiplication to Check a Quotient

$$\text{quotient} \cdot \text{divisor} + \text{remainder} = \text{dividend}$$

EXAMPLE 2

A Determine whether $2x + 4 + \frac{2}{x-5}$ is the quotient of $(2x^2 - 6x - 18) \div (x - 5)$.

SOLUTION

$$\begin{aligned}(2x + 4)(x - 5) + 2 &\stackrel{?}{=} 2x^2 - 6x - 18 && \text{quotient} \cdot \text{divisor} + \text{remainder} = \text{dividend} \\ 2x^2 - 10x + 4x - 20 + 2 &\stackrel{?}{=} 2x^2 - 6x - 18 && \text{Multiply.} \\ 2x^2 - 6x - 18 &= 2x^2 - 6x - 18 && \text{Simplify.}\end{aligned}$$

So $2x + 4 + \frac{2}{x-5}$ is the quotient of $(2x^2 - 6x - 18) \div (x - 5)$.

B Determine whether $3x - 5$ is the quotient of $(6x^2 - 2x - 15) \div (2x + 3)$.

SOLUTION

$$\begin{aligned}(3x - 5)(2x + 3) &\stackrel{?}{=} 6x^2 - 2x - 15 && \text{quotient} \cdot \text{divisor} = \text{dividend} \\ 6x^2 + 9x - 10x - 15 &\stackrel{?}{=} 6x^2 - 2x - 15 && \text{Multiply.} \\ 6x^2 - x - 15 &\neq 6x^2 - 2x - 15 && \text{Simplify.}\end{aligned}$$

So $3x - 5$ is not the quotient of $(6x^2 - 2x - 15) \div (2x + 3)$. ■

Dividing a Fourth-Degree Polynomial by a First-Degree Binomial

EXAMPLE 3

Divide $2x^4 - 4x + 7x^2 + 1$ by $x + 1$.

SOLUTION

Rewrite the dividend so that its terms are in order of decreasing degree, and write $0x^3$ for the missing x^3 term.

$$2x^4 - 4x + 7x^2 + 1 = 2x^4 + 0x^3 + 7x^2 - 4x + 1$$

Divide.

$$\begin{array}{r}
 2x^3 - 2x^2 + 9x - 13 \\
 x + 1 \overline{) 2x^4 + 0x^3 + 7x^2 - 4x + 1} \\
 \underline{-(2x^4 + 2x^3)} \\
 -2x^3 + 7x^2 \\
 \underline{-(-2x^3 - 2x^2)} \\
 9x^2 - 4x \\
 \underline{-(9x^2 + 9x)} \\
 -13x + 1 \\
 \underline{-(-13x - 13)} \\
 14
 \end{array}$$

The quotient is $2x^3 - 2x^2 + 9x - 13$. The remainder is 14.

$$(2x^4 - 4x + 7x^2 + 1) \div (x + 1) = 2x^3 - 2x^2 + 9x - 13 + \frac{14}{x + 1} \quad \blacksquare$$

► **Think About It** When you multiply a term in the quotient by the divisor, align the terms of the product with their like terms in the dividend.

Notice that in Example 3, the product $-(2x^4 + 2x^3)$ is written so that $2x^4$ is directly below $2x^4$ and $2x^3$ is directly below $0x^3$.

Determining Whether One Polynomial Is a Factor of Another Polynomial

EXAMPLE 4

Determine whether $a - 2$ is a factor of $a^3 - 7a + 6$.

SOLUTION

$$\begin{array}{r} a^2 + 2a - 3 \\ a - 2 \overline{) a^3 + 0a^2 - 7a + 6} \\ \underline{-(a^3 - 2a^2)} \\ 2a^2 - 7a \\ \underline{-(2a^2 - 4a)} \\ -3a + 6 \\ \underline{-(-3a + 6)} \\ 0 \end{array}$$

The remainder is 0, so $a - 2$ is a factor of $a^3 - 7a + 6$. ■

► **Think About It** When the remainder is 0, both the divisor and the quotient are factors of the dividend.

To check your answer, multiply $(a - 2)(a^2 + 2a - 3)$. Verify that the product is $a^3 - 7a + 6$.